

Digital Logic Design



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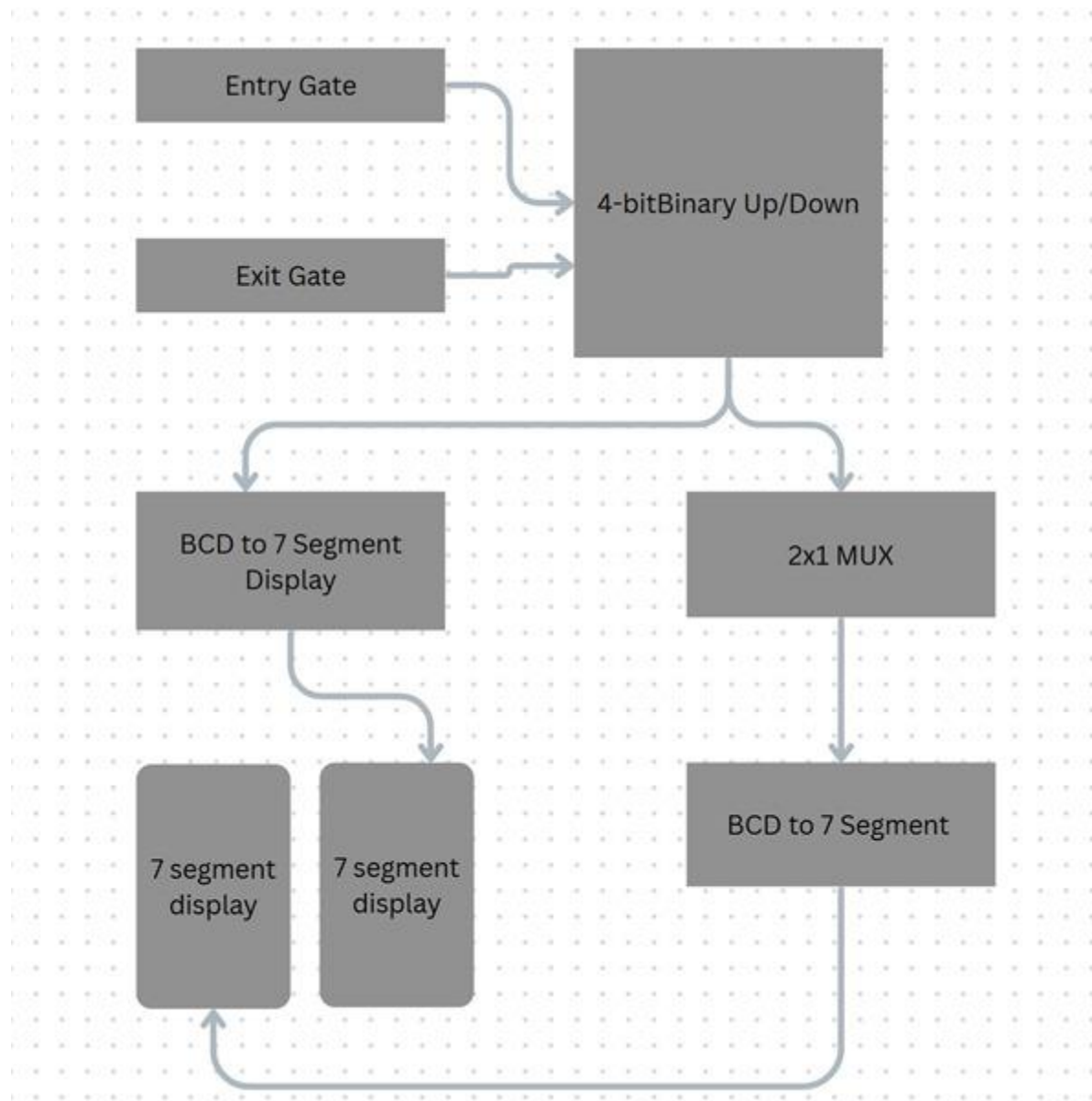
Project Title: Smart Parking Lot Management System

1. System Overview

The Smart Parking Lot Management System is a digital logic-based solution designed to automate the monitoring and management of a parking facility. Using basic logic gates and sequential circuits, the system:

- Detects the presence of vehicles at the entry and exit points.
- Allows vehicle entry only when parking space is available.
- Updates the internal binary counter representing the current number of parked vehicles.
- Prevents the entry of additional vehicles when the parking lot is full.
- Prevents vehicle exit when no cars are parked.
- Displays the current count of parked vehicles both in binary and on a 7-segment display.
- Includes an administrative manual override/reset feature to reinitialize the system.

2. Block Diagram



Admin Override/Reset Switch ---> System Reset & Control Logic

Explanation:

- The entry and exit sensors are implemented using SPST (single pole single throw) switches in simulation.
- Logic gates ensure that the vehicle can only enter if the count is less than the maximum (e.g., 15 slots).
- On each valid entry or exit, a rising-edge signal triggers the up/down counter.

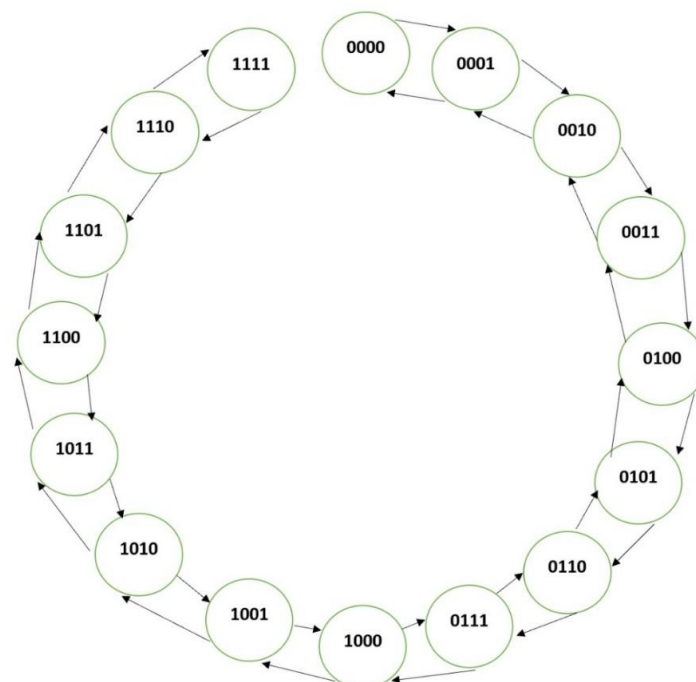
- The current count is displayed using a 4-bit binary output (for circuit debugging) and a 7-segment display (for user readability).
- A reset signal can bring the entire system back to its initial state.

3. Logic Components Used

- **Switches (Sensors):** Represent presence of vehicles at entry and exit points.
- **AND, OR, NOT Gates:** Used to create the logical conditions for valid entry/exit, full detection, and empty detection.
- **JK Flip-Flops (7473):** Used to build a 4-bit synchronous up/down counter. Toggled on valid entry or exit signals.
- **MUX IC 74157:** Used for data selection logic to simplify control between manual reset and sensor-driven operations.
- **BCD to 7-Segment Decoder (7448):** Converts binary counter output into a readable 7-segment display format.
- **LED Indicators:** Indicate parking full status, entry/exit gate operation, and system availability.
- **Manual Reset Button:** Admin control for system override or reset.

5. State Diagram

The counter maintains a state representing the number of parked vehicles:



- **Upward Transition:** Occurs on valid entry.
- **Downward Transition:** Occurs on valid exit.
- **Blocked Transitions:**
 - No increment when count is 15 (FULL).
 - No decrement when count is 0 (EMPTY).

4. Truth Tables

Entry Control Logic

Entry Sensor Parking Full Allow Entry Entry Gate Signal

0	X	0	0
1	0	1	1
1	1	0	0

Exit Control Logic

Exit Sensor Parking Empty Allow Exit Exit Gate Signal

0	X	0	0
1	0	1	1
1	1	0	0

To complete your report with detailed 4-bit Up/Down Counter Truth Table and Excitation Table (for JK Flip-Flops), we'll include the current state, next state, and the required J and K inputs for each flip-flop (Q_0 to Q_3). Since your counter:

- Counts up from 0000 to 1111, and stops at 1111.
- Counts down from 1111 to 0000, and stops at 0000.
- Does not roll over (e.g., does not go from 1111 $\rightarrow$ 0000 in up mode).

We assume it's a controlled counter, enabled only when entry or exit is valid.
Flip-Flop Excitation Table (JK)

Current (Q_n) Next (Q_{n+1}) J_n K_n

0	0	0	X
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Current (Q_n) Next (Q_{n+1}) J_n K_n

0 1 1 X

1 0 X 1

1 1 X 0

- **Current State (Q₃ Q₂ Q₁ Q₀)**
- **Next State for Up and Down**
- **Required JK inputs for each flip-flop (J₃–J₀, K₃–K₀)**
- **For both Up Counter and Down Counter**
- **Stops at 1111 in Up and at 0000 in Down**

Combined Truth and Excitation Table for 4-Bit Up/Down Counter (Stops at Limits)

Current State (Q₃ Q₂ Q₁ Q₀) Next State (UP) Next State (DOWN) J₃ K₃ J₂ K₂ J₁ K₁ J₀ K₀

0000	0001	0000	0	X	0	X	0	X	1	X
0001	0010	0000	0	X	0	X	1	X	X	1
0010	0011	0001	0	X	0	X	X	1	1	X
0011	0100	0010	0	X	1	X	X	1	X	1
0100	0101	0011	0	X	X	1	0	X	1	X
0101	0110	0100	0	X	X	1	1	X	X	1
0110	0111	0101	0	X	X	1	X	1	1	X
0111	1000	0110	1	X	0	X	0	X	X	1
1000	1001	0111	X	1	0	X	1	X	1	X
1001	1010	1000	X	1	0	X	X	1	X	1
1010	1011	1001	X	1	0	X	1	X	1	X
1011	1100	1010	X	1	1	X	0	X	X	1

Current State ($Q_3 Q_2 Q_1 Q_0$)	Next State (UP)	Next State (DOWN)	J_3	K_3	J_2	K_2	J_1	K_1	J_0	K_0
1100	1101	1011	X	1	X	1	1	X	1	X
1101	1110	1100	X	1	X	1	X	1	X	1
1110	1111	1101	X	1	X	1	1	X	1	X
1111	1111	1110	0	X	0	X	0	X	X	1

- X means "don't care" (either 0 or 1, depending on implementation).
- Flip-Flops are toggled only if:
 - $J = 1$ and $K = X \rightarrow$ Flip from 0 to 1
 - $J = X$ and $K = 1 \rightarrow$ Flip from 1 to 0
- The counter stops at 1111 while counting up and at 0000 while counting down (no rollover).

Final Simplified K-Map Expressions (Up Counter):

Flip-Flop J Expression K Expression

Q_0	1	1
Q_1	Q_0	Q_0
Q_2	$Q_1 \cdot Q_0$	$Q_1 \cdot Q_0$
Q_3	$Q_2 \cdot Q_1 \cdot Q_0$	$Q_2 \cdot Q_1 \cdot Q_0$

Down Counter

Just flip the conditions:

- Q_1 toggles when $Q_0 = 0$
- Q_2 toggles when $Q_1 = 0$ and $Q_0 = 0$
- Q_3 toggles when $Q_2 = 0, Q_1 = 0, Q_0 = 0$

Flip-Flop J (Down) K (Down)

Q_0	1	1
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Flip-Flop J (Down) K (Down)

$$Q_1 \quad Q_0'$$

$$Q_2 \quad Q_1' \cdot Q_0'$$

$$Q_3 \quad Q_2' \cdot Q_1' \cdot Q_0' \quad Q_2' \cdot Q_1' \cdot Q_0'$$

6. K-Map Simplification Examples

A. Full Detection Logic

Let A, B, C, D be outputs of the counter:

- Full condition: When A=1, B=1, C=1, D=1 (Binary 1111 = Decimal 15)
- Boolean Equation: **FULL** = $A \cdot B \cdot C \cdot D$

B. Empty Detection Logic

- Empty condition: A=0, B=0, C=0, D=0 (Binary 0000 = Decimal 0)
- Boolean Equation: **EMPTY** = $A' \cdot B' \cdot C' \cdot D'$

These equations are simplified from 4-variable K-Maps using SOP (Sum of Products) method.

7. Circuit Diagram

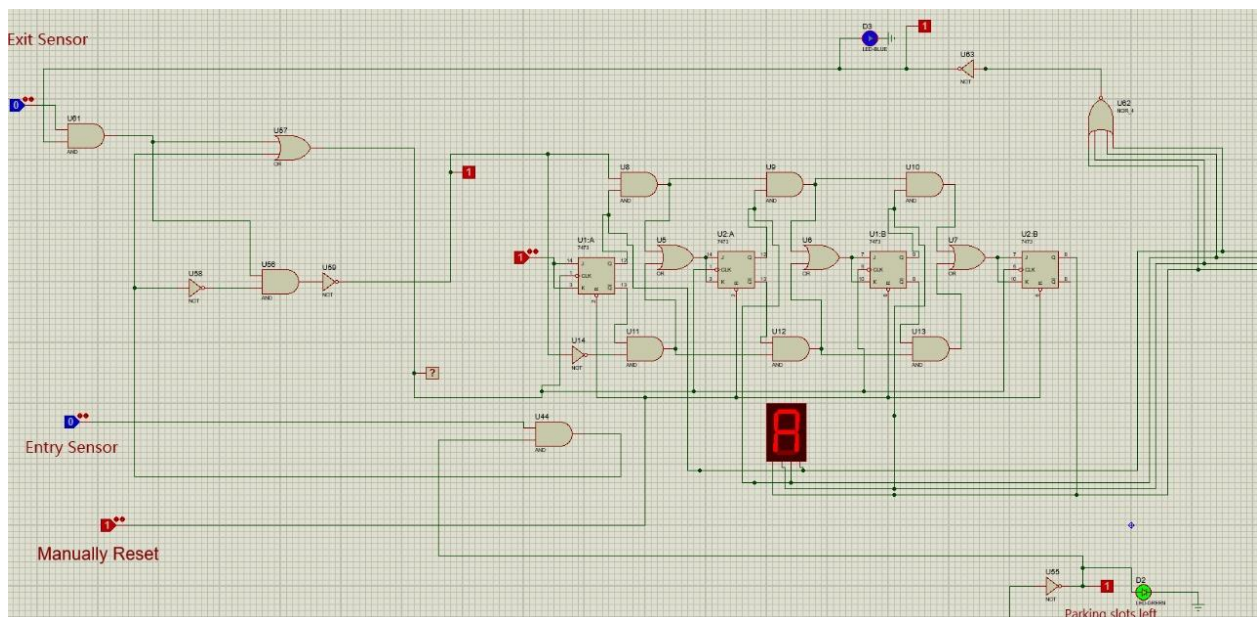


Figure: Proteus Simulation of Smart Parking Lot Controller

Explanation of Diagram:

- **Switches** at the bottom-left simulate the entry, exit, and manual reset sensors.
- The **AND/OR/NOT gates** form the decision logic for enabling the clock signal to the counter.
- The **flip-flops** configured as an up/down counter (likely 7473 ICs) track vehicle entries and exits.
- The **MUX IC 74157** handles conditional input routing.
- The **7448 Decoder** converts the 4-bit binary count into a 7-segment display format.
- The **7-segment display** shows the number of vehicles parked.
- The **LEDs** indicate gate status and parking availability (FULL signal).

8. Design Summary

Feature	Implementation Details
Entry/Exit Detection	Switches simulating sensors
Space Counter	4-bit Up/Down Counter using JK Flip-Flops (7473)
Display	Binary Output + 7-Segment Display (using 7448 BCD decoder)
Entry Control Logic	Prevents entry if parking is FULL
Exit Control Logic	Prevents exit if parking is EMPTY
Admin Override/Reset	Switch input resets the counter
MUX Logic Control	74157 MUX used for managing sensor vs manual logic
Circuit Simulated Using Proteus Design Suite	

Conclusion:

The Smart Parking Lot Management System is a digital logic-based solution that efficiently manages parking facilities by detecting vehicle presence at entry and exit points, updating parking counts in real-time, preventing overcrowding and unauthorized exits, and displaying current parking counts on a 7-segment display, with admin override/reset functionality for enhanced control.

